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COURSE:DSA0602-Data handling and visualization

1;**Perform Data Handling and Visualization on the Employee dataset using R. Analyze the dataset by handling data and generating different charts such as Histogram, Bar Chart, Pie Chart, Line Chart, Scatter Plot, Box Plot, Violin Plot, Density Plot, Area Chart, Correlation Matrix, and Pair Plot to understand employee characteristics** and trends.

Procedure

Open Google Colab (R Runtime) or RStudio.

Install and load the required libraries (ggplot2, dplyr, corrplot, and GGally).

Import the Employee.csv dataset.

Display the structure and summary of the dataset.

Check for missing values and duplicate records.

Convert categorical variables into factor data types if necessary.

Create the following visualizations:

Histogram

Bar Charts Pie

Chart

Line Chart

Area Chart

Scatter Plot

Box Plot

Violin Plot

Density Plot

Frequency Polygon

Dot Plot

Q-Q Plot

Stacked Bar Chart

Grouped Bar Chart

Correlation Matrix

Pair Plot

Analyze the generated graphs to understand employee distribution, demographics, experience, payment tiers, and leave status.

Interpret the observations obtained from the visualizations.

Education	JoiningYear	City	PaymentTie	Age	Gender	EverBenche	ExperienceI	LeaveOrNot
Bachelors	2017	Bangalore	3	34	Male	No	0	0
Bachelors	2013	Pune	1	28	Female	No	3	1
Bachelors	2014	New Delhi	3	38	Female	No	2	0
Masters	2016	Bangalore	3	27	Male	No	5	1
Masters	2017	Pune	3	24	Male	Yes	2	1
Bachelors	2016	Bangalore	3	22	Male	No	0	0
Bachelors	2015	New Delhi	3	38	Male	No	0	0
Bachelors	2016	Bangalore	3	34	Female	No	2	1
Bachelors	2016	Pune	3	23	Male	No	1	0
Masters	2017	New Delhi	2	37	Male	No	2	0

Load Libraries

library(ggplot2) library(dplyr)

Read Dataset employee <-

read.csv(file.choose())

Data Handling str(employee)

summary(employee)

colSums(is.na(employee))

employee <- distinct(employee)

Convert to Factors employee$Education <-

as.factor(employee$Education) employee$Gender <-

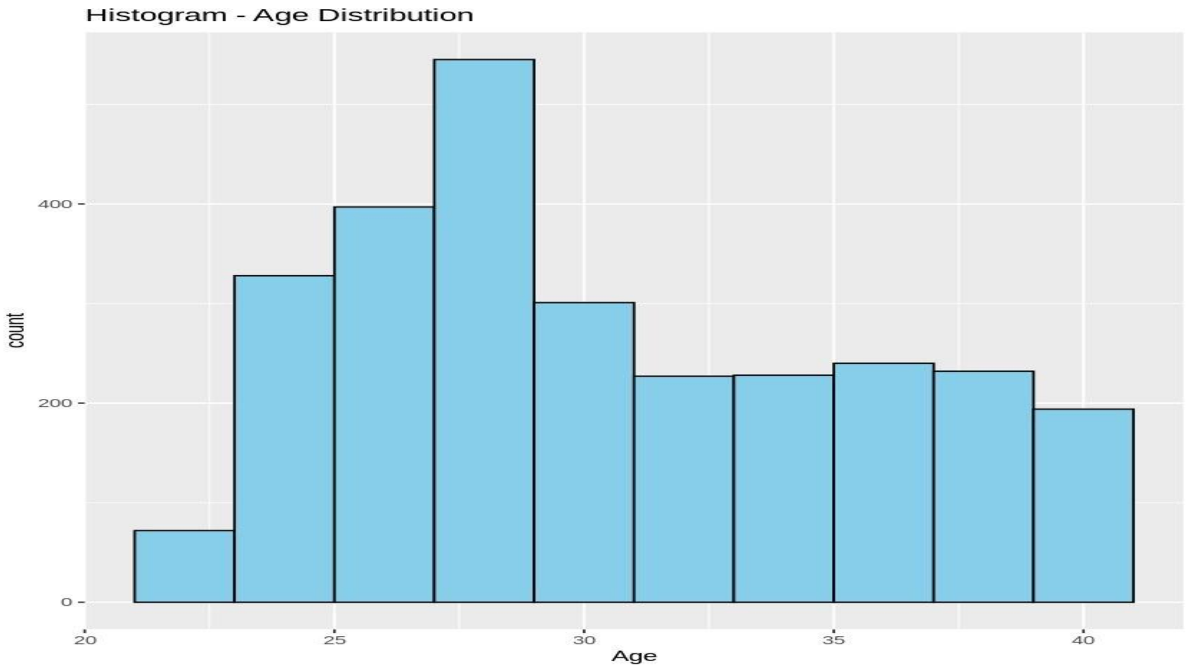
as.factor(employee$Gender) employee$City <-

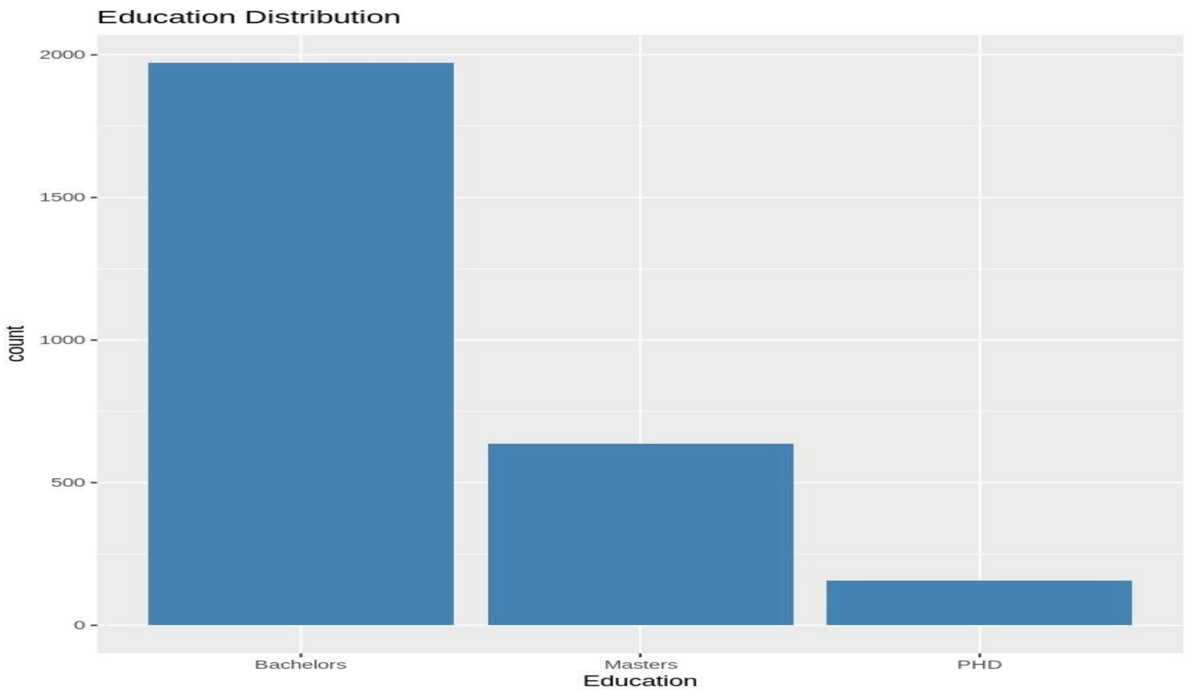
as.factor(employee$City) employee$LeaveOrNot <-

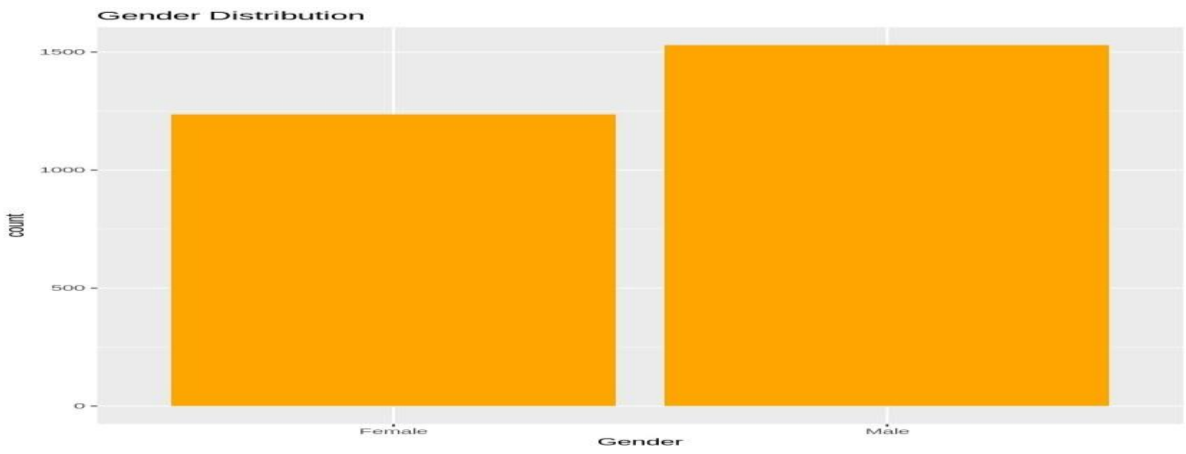
as.factor(employee$LeaveOrNot)

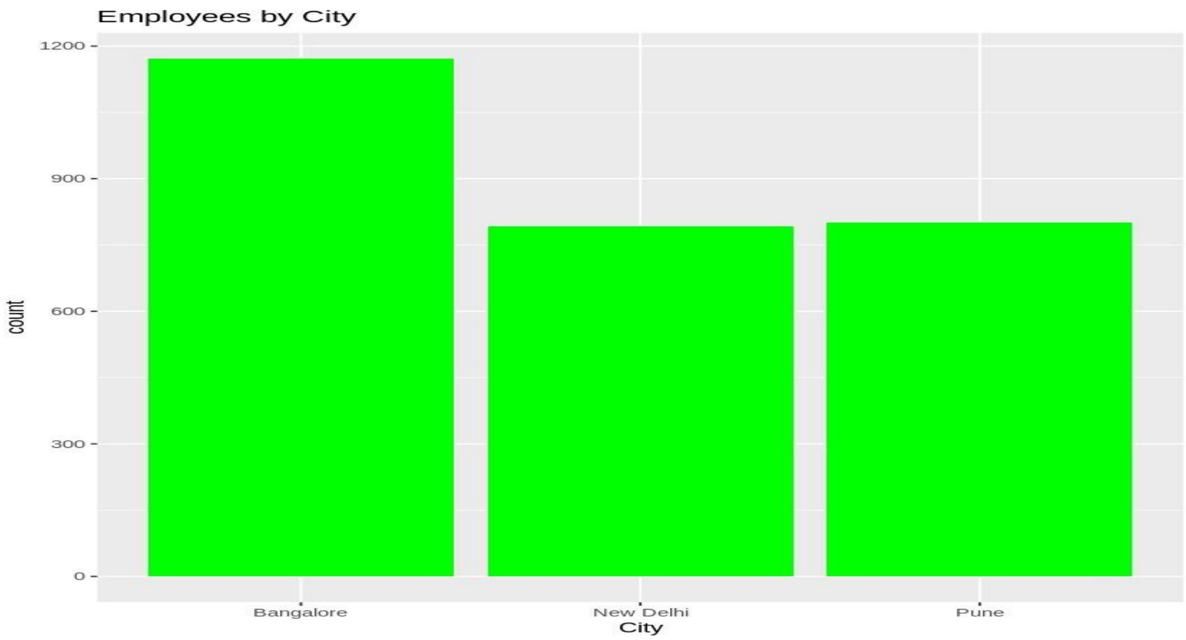
```
# Histogram ggplot(employee,aes(Age))+
geom_histogram(binwidth=2,fill="skyblue",color="black")+
ggtitle("Age Distribution")

# Bar Chart ggplot(employee,aes(Education))+
geom_bar(fill="steelblue")+
ggtitle("Education")

# Pie Chart leave <- employee %>%
count(LeaveOrNot)
ggplot(leave,aes(x="",y=n,fill=LeaveOrNot))+
geom_col(width=1)+ coord_polar("y")+
ggtitle("Leave Status")

# Scatter Plot
ggplot(employee,aes(Age,ExperienceInCurrentDomain,color=LeaveOrNot))+
geom_point(size=3)+ ggtitle("Age vs Experience")

# Box Plot ggplot(employee,aes(LeaveOrNot,Age,fill=LeaveOrNot))+
geom_boxplot()+
ggtitle("Age by Leave Status")

# Summary
table(employee$Education)
table(employee$Gender)
summary(employee)
```

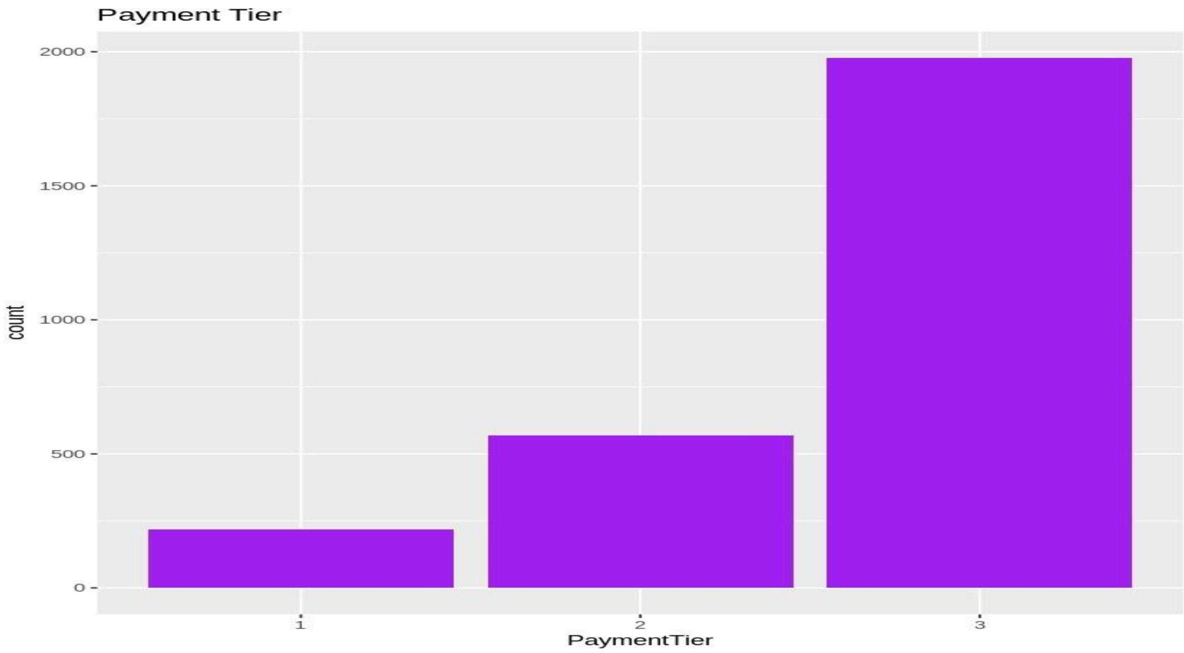
outcome:

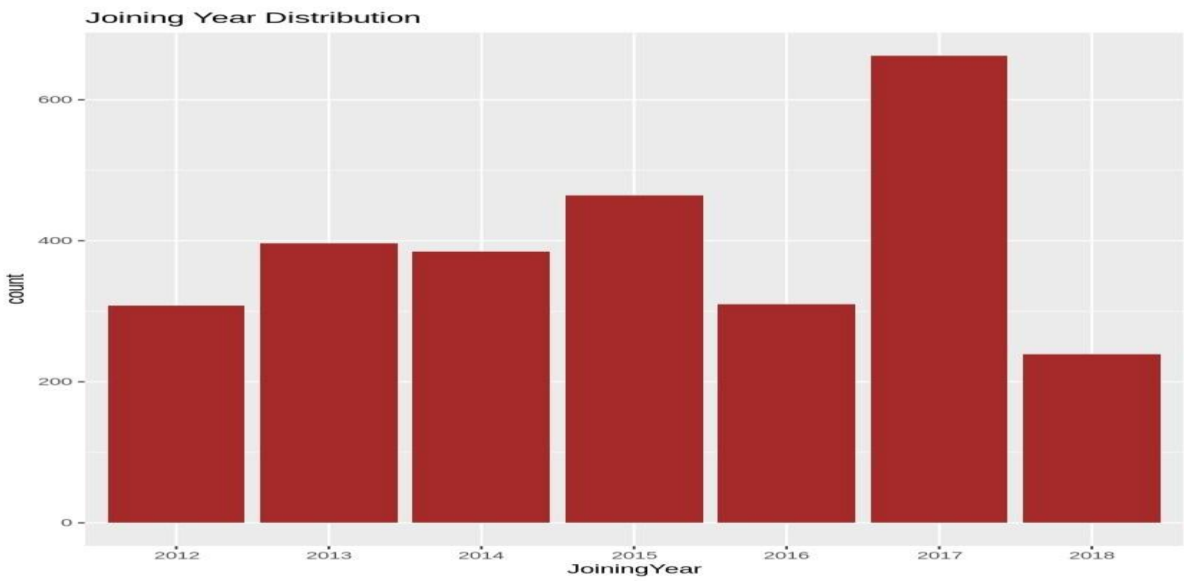

Histogram - Age Distribution
count
0
200
400
20
25
30
35
40
Age

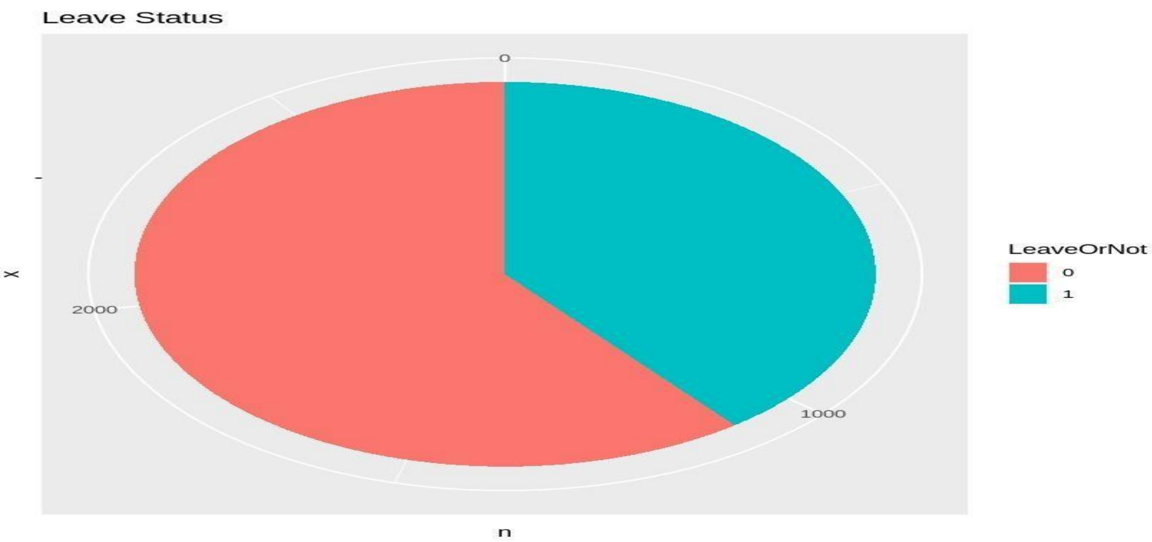

Education Distribution
count
0
500
1000
1500
2000
Bachelors
Masters
PHD
Education

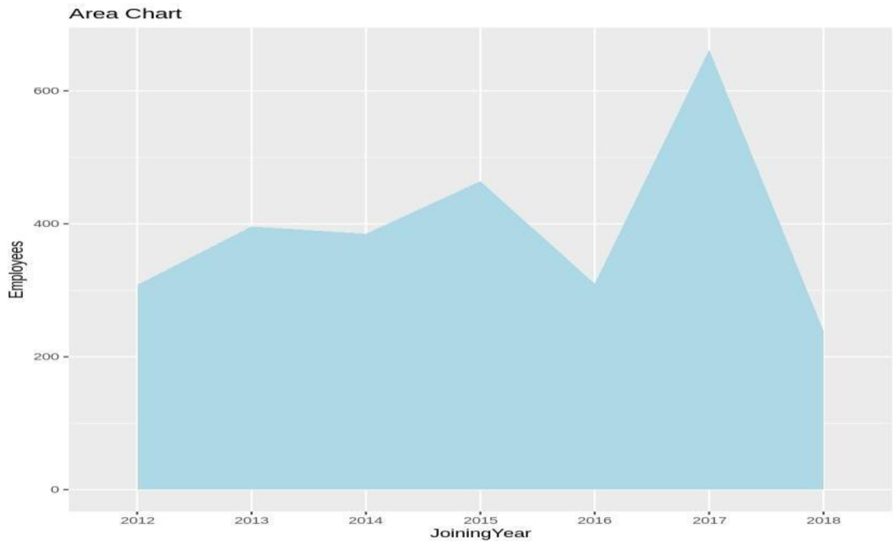

Gender Distribution
1500
1000
500
0
count
Female
Male
Gender

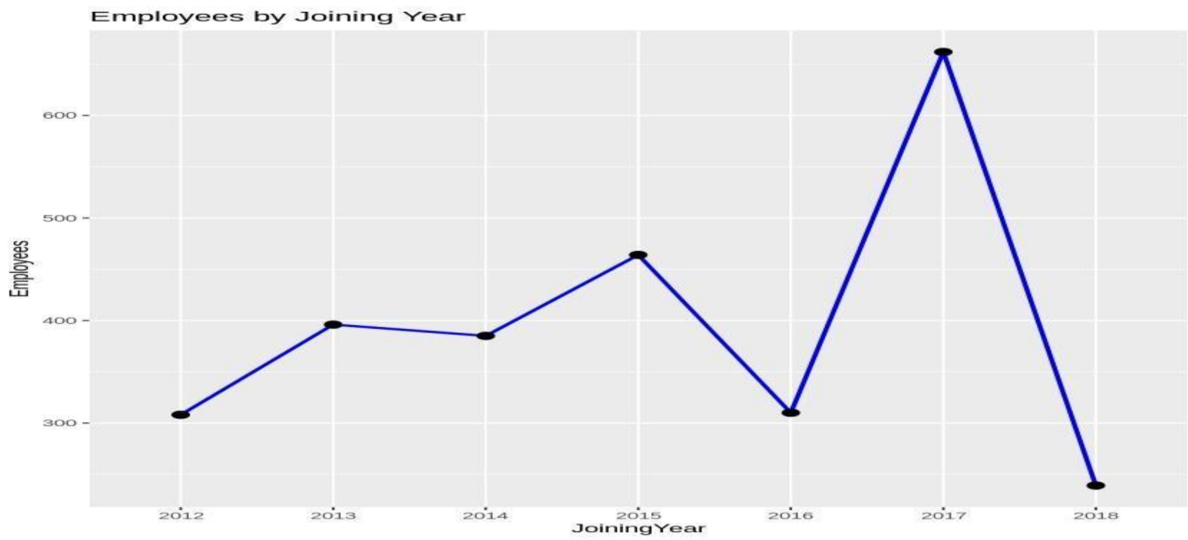

Employees by City
1200
900
600
300
0
count
Bangalore
New Delhi
Pune
City

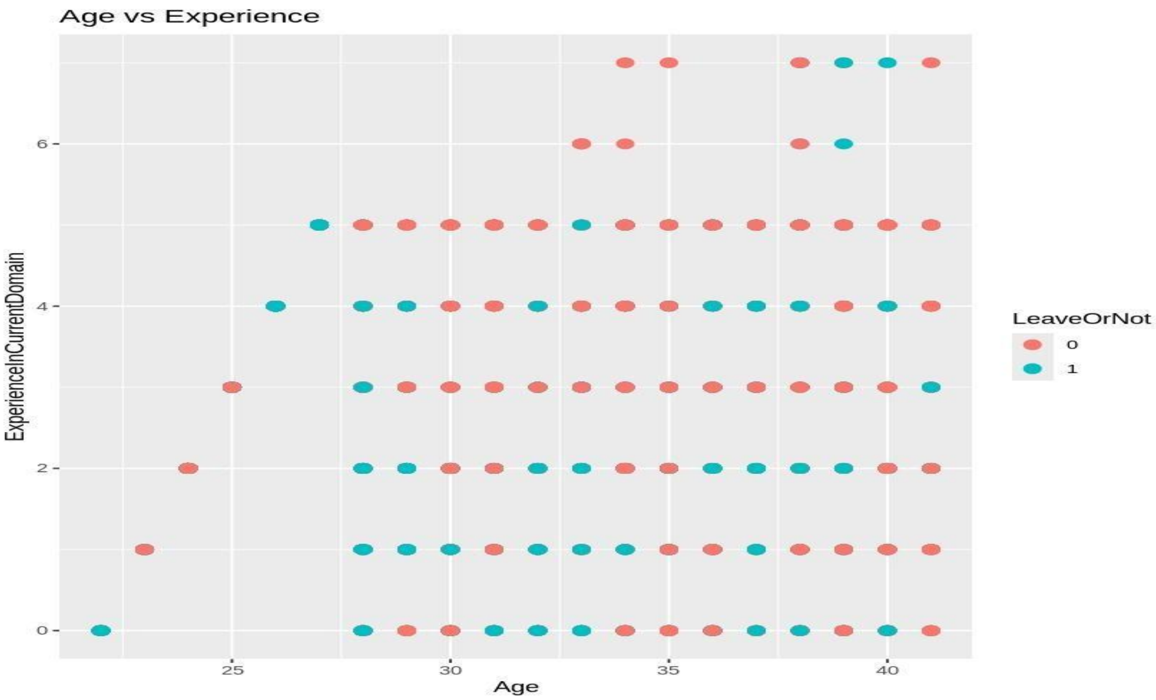

Payment Tier
count
2000
1500
1000
500
0
1
2
3
PaymentTier

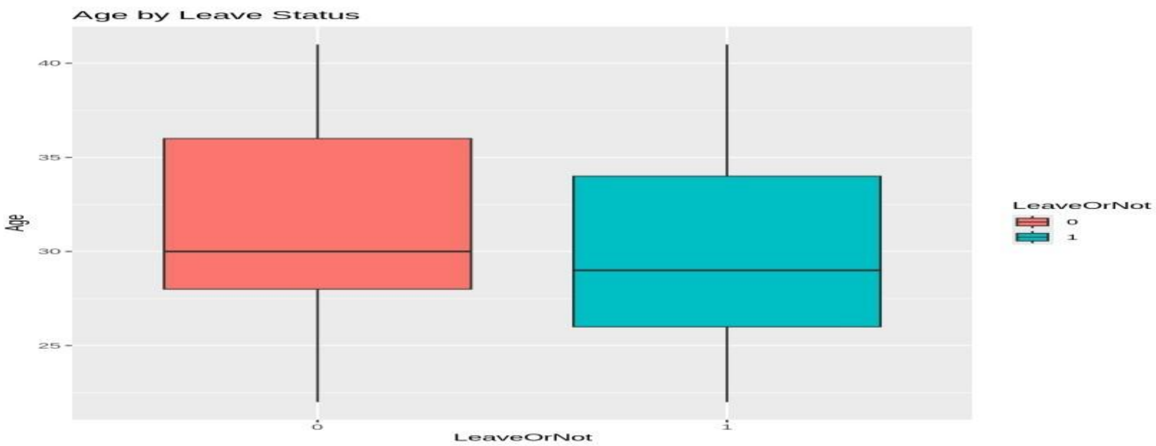

Joining Year Distribution
count
600
400
200
0
2012
2013
2014
2015
2016
2017
2018
JoiningYear

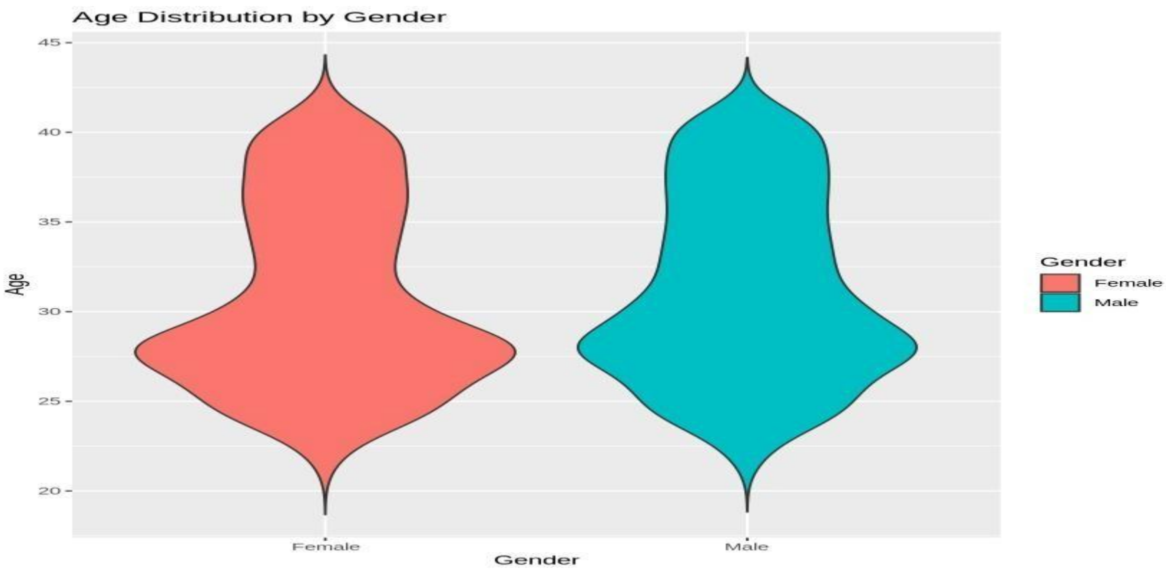

Leave Status
0
2000
1000
x
n
LeaveOrNot
0
1

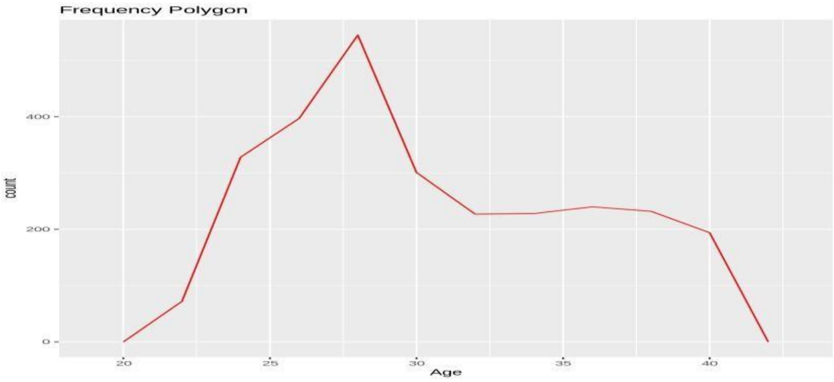

Area Chart
Employees
600
400
200
0
2012
2013
2014
2015
2016
2017
2018
JoiningYear


Employees by Joining Year
Employees
600
500
400
300
2012
2013
2014
2015
2016
2017
2018
JoiningYear


Age vs Experience
ExperienceInCurrentDomain
0
2
4
6
25
30
35
40
Age
LeaveOrNot
0
1


Age by Leave Status
Age
25
30
35
40
0
1
LeaveOrNot
LeaveOrNot
0
1


Age Distribution by Gender
Age
20
25
30
35
40
45
Female
Male
Gender
Gender
Female
Male



Result

The Employee dataset was successfully imported and analyzed using R. Data handling operations such as checking the dataset structure, summary statistics, and missing values were completed successfully. Various visualizations including Histogram, Bar Chart, Pie Chart, Line Chart, Area Chart, Scatter Plot, Box Plot, Violin Plot, Density Plot, Frequency Polygon, Dot Plot, Q-Q Plot, Stacked Bar Chart, Grouped Bar Chart, Correlation Matrix, and Pair Plot were generated successfully. The visualizations helped identify the distribution of employee age, education level, gender, city, payment tier, experience, and leave status, providing meaningful insights into the employee data.